

Global Contents

For

Final Detailed Project Report under Stage 4, for Revised Detailed Project Report and Verification of Executed Quantities / Items for Widening to 2 Lane for *Merangkong – Tamlu – Mon Road, Changtongya – Longleng Road, Chakabama – Zunheboto Road and Pfutsero – Phek Road* in State of Nagaland under SARDP Phase 1, is prepared and submitted in accordance with Para 18.6 of Terms of Reference. The Detailed Project Report consists of following Volumes:

Volume I:	Main Report
Appendix to Volume I:	Primary and Secondary Data
Volume II:	Design Report
Part I:	Road Features
Part II:	Bridges and Cross Drainage Structures
Appendix to Design Report:	Sub Soil Exploration Report
Volume III:	Materials Report
Volume IV:	Technical Specifications
Volume V:	Rate Analysis
Volume VI:	Cost Estimates
Volume VII:	Bill of Quantities
Volume VIII:	Drawings

Volume V: Rate Analysis for captioned project is presented in this Volume and is being submitted along with other, above detailed, Volumes.

Chapter 1

Introduction

1.1 The Consultancy Services

The Ministry of Road Transport and Highways, vide letter number NH-12018/03/2012/NG/SARDP-NE Dated 27th December 2013, has issued Letter of Acceptance to RITES Ltd. The Time Period for completion of activities under Terms of Reference is indicated as 9 months.

Performance Guarantee for subject works was submitted by RITES Ltd. on 20th January 2014. The Client officials has requested for clarification on submitted Performance Guarantee on 24th January 2014. The clarifications and amendments to Original Performance Guarantee were submitted by RITES Ltd. on 7th February 2014.

The Contract agreement has been signed on 17th April 2014. As sequel to signing of Contract Agreement, the Client officials have issued notice to consultant for proceeding with works on 19th May 2014. Accordingly consultancy works have been commenced and are in progress.

1.2 Project Roads

Four Project Roads under present scope of works are:

Table 1-1: Project Roads

S. No.	Name of State Road	Length as per ToR (Km)	Actual Length at Site (Km)	Location between (Lat./Long)	
				Start	End
1	Merangkong – Tamlu – Mon Road	100	98.1	26°36'/94°38'	26°43'/95°02'
2	Changtongya – Longleng Road	35	29.5	26°31'/94°40'	26°29'/94°50'
3	Chakabama – Zunheboto Road	128	121.2	25°39'/94°11'	26°02'/94°31'
4	Pfutsiro – Phek Road	66	65.3	25°34'/94°17'	25°41'/94°27'
	Total	329	314.1		

1.1.3. Merangkong – Tamlu – Mon (M T M) Road

Project Road starts from Junction with National Highway 61 near Km 209/500 from Kohima in Merangkong Village and terminates at km Stone No. 161 which is start point of Project 'Sewak' of Border Roads Organization (BRO) in Mon District. The Project Road traverses in Mokokchung, Longleng and Mon Districts of Nagaland. The Project Road passes through mountainous to steep terrain.



Start Point (Junction with NH-61) of M T M Road



End Point of M T M Road

1.2.2 Changtongya – Longleng (C L) Road

Project Road starts from Junction with National Highway 61 in Changtongya Village near Km 191 from Kohima. It connects Longleng District Head Quarters with National Highway. The Project Road traverses through Mokokchung and Longleng districts of Nagaland. The Locations of Start and End Points are detailed in photographs below:



Start Point, Junction with NH-61 in Changtongya



End Point in Longleng

During joint site visit held on 15 May 2014 with Nagaland PWD officials at Mokokchung, the physical end point of the project road was fixed near first traffic point from Changtongya. Length of Project Road is 29.530 Km.

1.2.3 Chakabama – Zunheboto (C Z) Road

Chakabama – Zunheboto Road is part of state Highway (approximate 170 km) from Mokokchung – Chakabama, connecting National Highways 61 (in North) and 29 (in South). The state Highway serves as connectivity to state capital, Zunheboto district headquarters and Nagaland University. The Project Road traverses through Phek and Zunheboto District of Nagaland State.

The Project Road starts from Junction with National Highway 29 in Chakabama Village near Km 27 from Kohima. It terminates at Junction after traversing Zunheboto town, near Assam Rifle establishment gate (km 122/250). The Project Road passes through mountainous to steep terrain.



Start Point, Junction with NH-29 in Chakabama



End Point in Zunheboto

1.2.4 Pfutsero – Phek (P P) Road

Project Road starts from Pfutsero, at junction with National Highway 29 near Km 66/500 from Kohima and Ends in Phek at junction in front of Gate of camp area of 'Assam Rifle'. The Project Road would provide 2-lane connectivity to Phek district Headquarters from NH-29 through interior and remote villages.



Start Point, Junction with NH-29 in Pfutsero



End Point in Phek

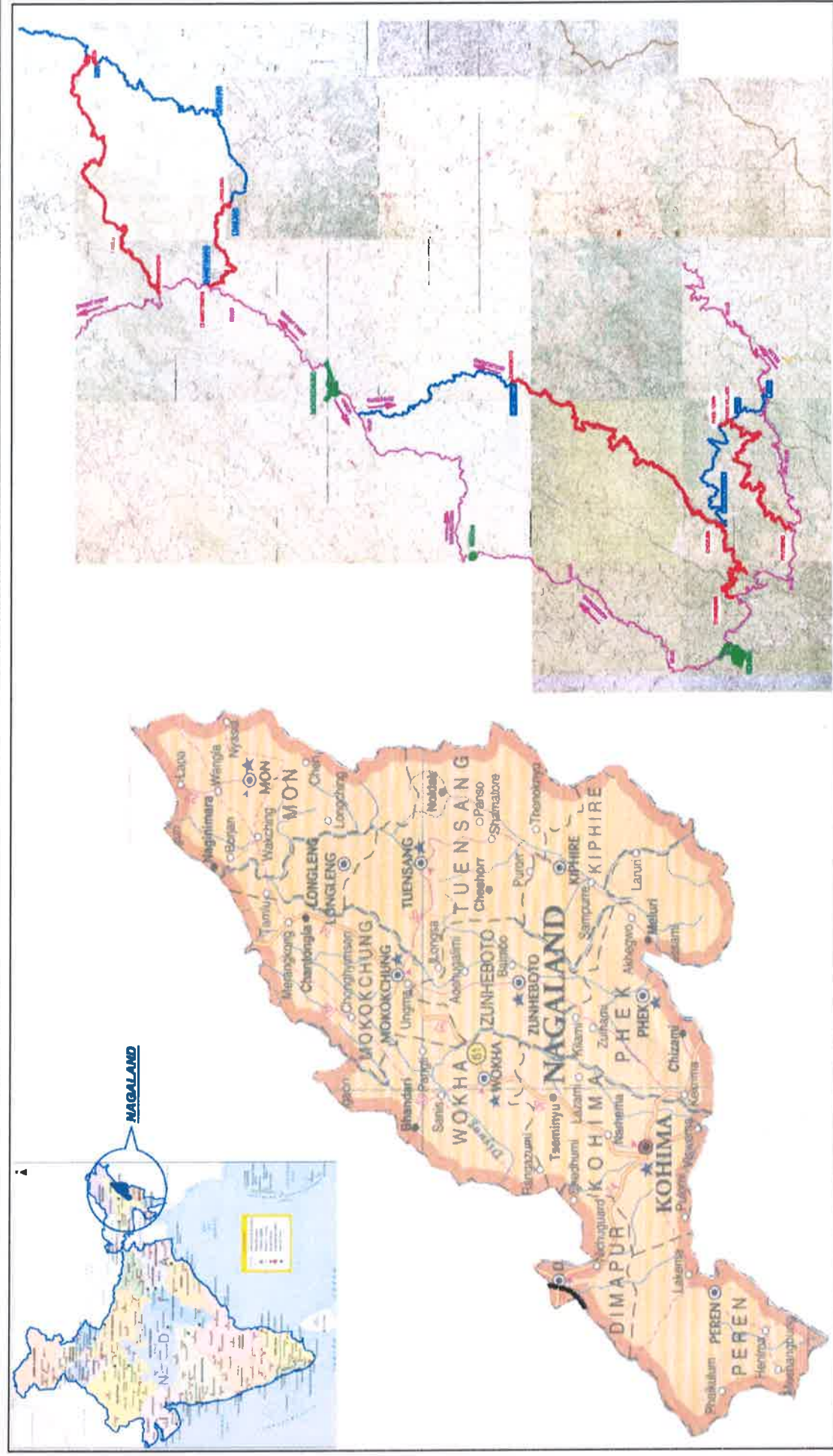


Figure 1-1: Key Plan of Project Roads



Chapter 3

Rate Analysis



INPUT RATES OF PLANT, MACHINERY, LABOUR AND MATERIALS
(A) Usage Rates of Plant and Machinery

Sl. No.	Description of Machine	Activity	Output of Machine	Output	Unit	Rate (Rs.)
P&M-001	Air Compressor	General Purpose	capacity In cfm	170/250	hour	353.10
P&M-002	Batching and Mixing Plant (a) 30 cum capacity	Concrete Mixing	cum/hour	20	hour	2,361.70
P&M-003	Batching and Mixing Plant (b) 15 - 20 cum capacity	Concrete Mixing	cum/hour	13	hour	1,969.00
P&M-004	Bitumen Pressure Distributor	Applying bitumen tack coat	sqm/hour	1750	hour	1,135.20
P&M-005	Bitumen Boiler oil fired	Bitumen Spraying	capacity In litre	1500	hour	347.60
P&M-006	Concrete Paver Finisher with 40 HP Motor	Paving of concrete surface	cum / hour	20	hour	3,418.80
P&M-007	Concrete Pump of 45 & 30 cum capacity	Pumping of concrete	cum / hour	33 / 22	hour	460.90
P&M-008	Concrete Bucket	For Pouring concrete	capacity In cum	1	hour	20.90
P&M-009	Concrete Mixer (a) 0.4/0.28 cum	Concrete Mixing	cum/hour	2.5	hour	474.10
P&M-010	Concrete Mixer (b) 1 cum	Concrete Mixing	cum/hour	7.5	hour	795.30
P&M-011	Crane (a) 80 tonnes	Lifting Purpose			hour	1,936.00
P&M-012	Crane (b) 35 tonnes	Lifting Purpose			hour	1,289.20
P&M-013	Crane (c) 3 tonnes	Lifting Purpose			hour	440.00
P&M-014	Dozer D - 80 - A 12	Spreading /Cutting / Clearing	cum/hour	300/150/250	hour	3,936.90
P&M-015	Dozer D - 50 - A 15	Spreading /Cutting / Clearing	cum/hour	200/120/150	hour	2,778.60
P&M-016	Emulsion Pressure Distributor	Applying emulsion tack coat	sqm/hour	1750	hour	885.50
P&M-017	Front End loader 1 cum bucket capacity	Soil loading / Aggregate loading	cum/hour	60 /25	hour	1,016.40
P&M-018	Generator (a) 125 KVA	Generation of electric Energy	KVA	100	hour	773.30
P&M-019	Generator (b) 63 KVA	Generation of electric Energy	KVA	50	hour	436.70
P&M-020	GSB Plant 50 cum	Producing GSB	cum/hour	40	hour	1,150.60
P&M-021	Hotmix Plant - 120 TPH capacity	DBM/BM/SDC/ Premix	cum/hour	40	hour	24,777.50
P&M-022	Hotmix Plant - 100 TPH capacity	DBM/BM/SDC/ Premix	cum/hour	30	hour	22,685.30
P&M-023	Hotmix Plant - 60 to 90 TPH capacity	DBM/BM/SDC/ Premix	cum/hour	25	hour	18,141.20
P&M-024	Hotmix Plant - 40 to 60 TPH capacity	DBM/BM/SDC/ Premix	cum/hour	17	hour	14,525.50
P&M-025	Hydraulic Chip Spreader	Surface Dressing	sqm/hour	1500	hour	2,789.60
P&M-026	Hydraulic Excavator of 1 cum bucket	Soil Ordinary/ Soil Marshy/ Soil Unsuitable	cum/hour	60 /60 /60	hour	1,575.20
P&M-027	Integrated Stone Crusher 100TPH	Crushing of Spalls	TPH	100	hour	9,171.80
P&M-028	Integrated Stone Crusher 200 HP	Crushing of Spalls	TPH	200	hour	18,378.80
P&M-029	Kerb Casting Machine	Kerb Making	Rm/hour	80	hour	382.80
P&M-030	Mastic Cooker	Mastic Wearing coat	capacity In tonne	1	hour	62.84
P&M-031	Mechanical Broom Hydraulic	Surface Cleaning	sqm/hour	1250	hour	361.90
P&M-032	Motor Grader 3.35 mtr blade	Clearing /Spreading /GSB /WBM	cum/hour	200/200/50/50	hour	2,536.60
P&M-033	Mobile slurry seal equipment	Mixing and laying slurry seal	sqm/hour	2700	hour	1,067.00
P&M-034	Paver Finisher Hydrostatic with sensor control 100 TPH	Paving of DBM/ BM/SDC/ Premix	cum/hour	40	hour	2,965.60
P&M-035	Paver Finisher Mechanical 100 TPH	Paving of WMM /Paving of DLC	cum/hour	40/30	hour	1,474.00
P&M-036	Piling Rig with Bantonite Pump	0.75 m dia to 1.2 m dia Boring attachment	Rm/hour	2 to 3	hour	5,783.80
P&M-037	Pneumatic Road Roller	Rolling of Asphalt Surface	cum/hour	25	hour	1,378.30
P&M-038	Pneumatic Sinking Plant	Pneumatic Sinking of wells	cum/hour	1.5 to 2.00	hour	4,414.30
P&M-039	Pot Hole Repair Machine	Repair of potholes	cum/hour	4	hour	960.30
P&M-040	Prestressing Jack with Pump & access	Stressing of steel wires/stands			hour	144.10
P&M-041	Ripper	Scarifying	cum/hour	60	hour	46.20
P&M-042	Rotavator	Scarifying	cum/hour	25	hour	50.60
P&M-043	Road marking machine	Road marking	sqm/hour	100	hour	104.50
P&M-044	Smooth Wheeled Roller 8 tonne	Soil /BM Compaction	cum/hour	70/25	hour	555.50
P&M-045	Tandem Road Roller	Rolling of Asphalt Surface	cum/hour	30	hour	856.90
P&M-046	Tipper - 5 cum	Transportation of soil, GSB, WMM, Hotmix etc.	Capacity In cum	5.5	km	191.40
P&M-047	Tipper - 5 cum	Transportation of soil, GSB, WMM, Hotmix etc.	Capacity In cum	5.5	tonne.km	30.25
P&M-048	Tipper - 5 cum	Transportation of soil, GSB, WMM, Hotmix etc.	Capacity In cum	5.5	hour	701.80
P&M-049	Transit Mixer 4.0/4.5 cum	Transportation of Concrete Mix to site	cum/hour	4.5	hour	1,095.60

INPUT RATES OF PLANT, MACHINERY, LABOUR AND MATERIALS

P&M-050	Transit Mixer 4.0/4.5 cum	Transportation of Concrete Mix to site	cum/hour	4.5	tonne.km	42.35
P&M-051	Transit Mixer 3.0 cum	Transportation of Concrete Mix to site	cum/hour	3	hour	99.00
P&M-052	Transit Mixer 3.0 cum	Transportation of Concrete Mix to site	cum/hour	3	tonne.km	30.80
P&M-053	Tractor	Pulling	capacity in HP	50	hour	402.60
P&M-054	Tractor with Rotavator	Rate of Tractor + Rotavator			hour	422.40
P&M-055	Tractor with Ripper	Rate of Tractor 6+ Ripper			hour	433.40
P&M-056	Truck 5.5 cum per 10 tonnes	Material Transport	capacity / cum	4.5	km	149.60
P&M-057	Truck 5.5 cum per 10 tonnes	Material Transport	capacity / cum	4.5	hour	748.00
P&M-058	Truck 5.5 cum per 10 tonnes	Material Transport	capacity / cum	4.5	tonne.km	5.04
P&M-059	Vibratory Roller 8 tonne	Earth or soil / GSB / WBM	cum/hour	100/60/60	hour	1,103.30
P&M-060	Water Tanker	Water Transport	capacity In KL	6	hour	748.00
P&M-061	Water Tanker	Water Transport	capacity In KL	6	km	24.93
P&M-062	Wet Mix Plant 60 TPH	Wet Mix	cum/hour	25	hour	1,335.40
Sl. No.	Description of Machine				Unit	Rate (Rs.)
P&M-063	Air compressor with pneumatic chisel attachment for cutting hard clay.				hour	315.32
P&M-064	Batch type cold mixing plant 100-120 TPH capacity producing an average output of 75 tonne per hour				hour	4,102.56
P&M-065	Belt conveyor system				hour	683.76
P&M-066	Boat to carry atleast 20 persons				hour	371.80
P&M-067	Cement concrete batch mix plant @ 175 cum per hour (effective output)				hour	4,102.56
P&M-068	Cement concrete batch mix plant @ 75 cum per hour				hour	3,008.78
P&M-069	Cold milling machine @ 20 cum per hour				hour	2,461.31
P&M-070	Crane 5 tonne capacity				hour	579.81
P&M-071	Crane 10 tonne capacity				hour	687.23
P&M-072	Crane 15 tonne capacity				hour	816.59
P&M-073	Crane 20 tonne capacity				hour	947.10
P&M-074	Crane 40 T capacity				hour	1,076.46
P&M-075	Crane with grab 0.75 cum capacity				hour	744.15
P&M-076	Compressor with gunting equipment along with accessories				hour	1,367.52
P&M-077	Drum mix plant for cold mixes of appropriate capacity but not less than 75 tonnes/hour.				hour	711.48
P&M-078	Epoxy Injection gun				hour	443.30
P&M-079	Generator 33 KVA				hour	585.75
P&M-080	Generator 100 KVA				hour	781.28
P&M-081	Generator 250 KVA				hour	1,758.90
P&M-082	Induction, delinduction and erection of plant and equipment including all components and accessories for pneumatic method of well sinking.				hour	4,030.29
P&M-083	Joint Cutting Machine with 2-3 blades (for rigid pavement)				hour	488.40
P&M-084	Jack for Lifting 40 tonne lifting capacity.				day	519.20
P&M-085	Piling rig including double acting pile driving hammer (Hydraulic rig)				hrs	205.59
P&M-086	Plate compactor				hour	205.59
P&M-087	Snow blower equipment 140 HP @ 600 cum per hour				hour	820.05
P&M-088	Texturing machine (for rigid pavement)				hour	1,628.00
P&M-089	Truck Tractor 30 tonne capacity				hour	952.88
P&M-090	Truck Tractor 30 tonne capacity				t.km	181.50
P&M-091	Tunnel Boring machine				hour	2,344.65
P&M-092	Vibrating Pile driving hammer complete with power unit and accessories				hour	1,145.87
P&M-093	Wet Mix Plant 100 TPH				hour	2,032.80
P&M-094	Wet Mix Plant 75 TPH				hour	1,407.12

(B) Labour

Sl. No.	Description of Labour	Unit	Rate (Rs.)
L-01	Blacksmith (IInd class)	day	357.50
L-02	Blacksmith (1st class)/ Welder/ Plumber/ Electrician	day	396.00
L-03	Blaster (Stone cutter)	day	396.00
L-04	Carpenter I Class	day	396.00
L-05	Chiseller (Head Mazdoor)	day	396.00
L-06	Driller (Jumper)	day	396.00
L-07	Driver	day	467.50
L-08	Fitter	day	396.00
L-09	Mail	day	357.50
L-10	Mason (IInd class)	day	357.50
L-11	Mason (1st class)	day	396.00
L-12	Mate / Supervisor	day	440.00
L-13	Mazdoor	day	302.50
L-14	Mazdoor/Dresser (Semi Skilled)	day	330.00
L-15	Mazdoor/Dresser/Sinker (Skilled)	day	357.50
L-16	Medical Officer	day	935.00
L-17	Operator (grouting)	day	396.00
L-18	Painter I class	day	396.00
L-19	Para medical personnel	day	715.00

INPUT RATES OF PLANT, MACHINERY, LABOUR AND MATERIALS

Lead Table						
Sl. No.	Description	From	To	Lead		
				Surface Road	Katcha Track	Total
1	Earthwork	Borrow Area	Site	0.00	0.00	0.00
2	Finished Materials	HMP Plant / Batching Plant / Crushing Plant	Site	10.00	0.00	10.00
(C) Materials						
Sl. No.	Description	Unit	Rate	Carriage (Rs)	Rate (Rs.)	
M-001	Stone Boulder of size 150 mm and below at Crusher Plant	cum	2,079.00	0.00	2,286.90	
M-002	Supply of quarried stone 150 - 200 mm size for Hand Broken at site	cum	1,975.05	0.00	2,172.56	
M-003	Boulder with minimum size of 300 mm for Pitching at Site	cum	1,917.00	0.00	2,108.70	
M-004	Coarse sand at Mixing Plant	cum	5,327.00	0.00	5,859.70	
M-005	Coarse sand at Site	cum	5,918.42	0.00	6,510.26	
M-006	Fine sand at Site	cum	5,085.17	0.00	5,593.69	
M-007	Moorum at Site	cum	1,639.36	0.00	1,803.30	
M-008	Gravel/Quarry spall at Site	cum	2,782.50	0.00	3,060.75	
M-009	Granular Material or hard murrum for GSB works at site	cum	1,803.29	0.00	1,983.62	
M-010	Granular Material or hard murrum for GSB works at Mixing Plant	cum	1,803.29	0.00	1,983.62	
M-011	Fly ash conforming to IS: 3812 (Part II & I) at HMP Plant / Batching Plant / Crushing Plant	cum	6,190.45	0.00	6,809.50	
M-012	Filter media/Filter Material as per Table 300-3 (MoRT&H Specification)	cum	1,741.00	0.00	1,915.10	
Sl. No.	Description	Unit	Rate at Plant (HMP/Batching) (Rs.)		Rate at Site (Rs.)	
M-013	Close graded Granular sub-base Material 53 mm to 9.5 mm	cum	1655.00		1,820.50	
M-014	Close graded Granular sub-base Material 37.5 mm to 9.5 mm	cum	1675.00		1,842.50	
M-015	Close graded Granular sub-base Material 26.5 mm to 9.5 mm	cum	1711.00		1,882.10	
M-016	Close graded Granular sub-base Material 9.5 mm to 4.75 mm	cum	1736.00		1,909.60	
M-017	Close graded Granular sub-base Material 9.5 mm to 2.36 mm	cum	1741.00		1,915.10	
M-018	Close graded Granular sub-base Material 4.75mm to 2.36 mm	cum	1746.00		1,920.60	
M-019	Close graded Granular sub-base Material 4.75mm to 75 micron mm	cum	1749.00		1,923.90	
M-020	Close graded Granular sub-base Material 2.36 mm	cum	1751.00		1,926.10	
M-021	Stone crusher dust finer than 3mm with not more than 10% passing 0.075 sieve	cum	1639.36		1,803.30	
M-022	Coarse graded Granular sub-base Material 2.36 mm & below	cum	2561.27		2,817.40	
M-023	Coarse graded Granular sub-base Material 4.75mm to 75 micron mm	cum	2556.27		2,811.90	
M-024	Coarse graded Granular sub-base Material 4.75 mm to 2.36 mm	cum	2551.27		2,806.40	
M-025	Coarse graded Granular sub-base Material 9.5 mm to 4.75 mm	cum	2536.27		2,789.90	
M-026	Coarse graded Granular sub-base Material 26.5 mm to 4.75 mm	cum	2511.27		2,762.40	
M-027	Coarse graded Granular sub-base Material 26.5 mm to 9.5 mm	cum	2506.27		2,756.90	
M-028	Coarse graded Granular sub-base Material 37.5 mm to 9.5 mm	cum	2481.27		2,729.40	
M-029	Coarse graded Granular sub-base Material 53 mm to 26.5mm	cum	2431.27		2,674.40	
M-030	Aggregates below 5.6 mm	cum	2025.00		2,227.50	
M-031	Aggregates 22.4 mm to 2.36 mm	cum	1819.00		2,000.90	
M-032	Aggregates 22.4 mm to 5.6 mm	cum	1816.00		1,997.60	
M-033	Aggregates 45 mm to 2.8 mm	cum	1763.00		1,939.30	
M-034	Aggregates 45 mm to 22.4 mm	cum	1741.00		1,915.10	
M-035	Aggregates 53 mm to 2.8 mm	cum	1773.00		1,950.30	
M-036	Aggregates 53 mm to 22.4 mm	cum	1761.00		1,937.10	
M-037	Aggregates 63 mm to 2.8 mm	cum	1758.00		1,933.80	
M-038	Aggregates 63 mm to 45 mm	cum	1589.00		1,747.90	
M-039	Aggregates 90 mm to 45 mm	cum	1559.00		1,714.90	
M-040	Aggregates 10 mm to 5 mm	cum	1999.00		2,198.90	
M-041	Aggregates 11.2 mm to 0.09 mm	cum	1997.00		2,196.70	
M-042	Aggregates 13.2 mm to 0.09 mm	cum	1995.00		2,194.50	
M-043	Aggregates 13.2 mm to 5.6 mm	cum	1990.00		2,189.00	
M-044	Aggregates 13.2 mm to 10 mm	cum	1970.00		2,167.00	
M-045	Aggregates 20 mm to 10 mm	cum	1817.00		1,998.70	
M-046	Aggregates 25 mm to 10 mm	cum	1812.00		1,993.20	
M-047	Aggregates 19 mm to 6 mm	cum	1829.00		2,011.90	
M-048	Aggregates 37.5 mm to 19 mm	cum	1782.00		1,960.20	
M-049	Aggregates 37.5 mm to 25 mm	cum	1777.00		1,954.70	
M-050	Aggregates 6 mm nominal size	cum	2010.00		2,211.00	
M-051	Aggregates 10 mm nominal size	cum	1980.00		2,178.00	
M-052	Aggregates 13.2/12.5 mm nominal size	cum	1970.00		2,167.00	
M-053	Aggregates 20 mm nominal size	cum	1817.00		1,998.70	
M-054	Aggregates 25 mm nominal size	cum	1807.00		1,987.70	
M-055	Aggregates 40 mm nominal size	cum	1741.00		1,915.10	

INPUT RATES OF PLANT, MACHINERY, LABOUR AND MATERIALS

Sl. No.	Description	Unit	Rate (Rs.)
M-056	AC pipe 100 mm dia	metre	381.15
M-057	Acrylic polymer bonding coat	litre	473.55
M-058	Aluminium Paint	litre	343.20
M-059	Aluminium alloy plate 2mm Thick	sqm	984.50
M-060	Aluminium alloy/galvanised steel	tonne	1,60,854.38
M-061	Aluminium sheeting fixed with encapsulated lens type reflective sheeting Including 2% towards lettering, cost of angle iron, cost of drilling holes, nuts, bolts etc.and signs as applicable	sqm	10,781.10
M-062	Aluminium studs 100 x 100 mm fitted with lense reflectors	nos	1,113.20
M-063	Barbed wire	kg	457.38
M-064	Bearing (Cost of parts)	nos	94,915.43
M-065	Bearing (Cast steel rocker bearing assembly of 250 tonne)	nos	1,00,273.31
M-066	Bearing (Elastomeric bearing assembly consisting of 7 internal layers of elastomer bonded to 6 nos. Internal reinforcing steel laminates by the process of vulcanisation,)	nos	95,711.00
M-067	Bearing (Forged steel roller bearing of 250 tonne	nos	93,984.33
M-068	Bearing (Pot type bearing assembly consisting of a metal piston supported by a disc, PTFE pads providing sliding surfaces against stainless steel mating together with cast steel assemblies/ fabricated structural steel assemblies duly painted with all components	nos	30,800.00
M-069	Bearing (PTFE sliding plate bearing assembly of 80 tonnes)	nos	10,521.50
M-070	Bearing (Supply of sliding plate bearing of 80 tonne)	nos	10,521.50
M-071	Bentonite	kg	197.51
M-072	Binding wire	kg	79.75
M-073	Bitumen (Cationic Emulsion)	tonne	81,331.48
M-074	Bitumen (50-70 grade)	tonne	35,934
M-077	Bitumen (emulsion)	tonne	31,428
M-078	Bitumen (modified graded)	tonne	39,455
M-079	Brick	each	15.69
M-080	C.I.shoes for the pile	kg	62.37
M-081	Cement	tonne	8,629.20
M-082	Thermo Mechanically Treated bars (HYSD/TMT Bars)	tonne	59,859.20
M-083	Collar for joints 300 mm dia	nos	959.81
M-084	Compressible Fibre Board(20mm thick)	sqm	1,029.11
M-085	Connectors/ Staples	each	65.84
M-086	Copper Plate(12m long x 250mmwide)	kg	43,340.00
M-087	Corrosion resistant Structural steel	tonne	93,500.00
M-088	Corrugated sheet, 3 mm thick, "Thrie" beam section railing	kg	86.35
M-089	Credit for excavated rock found suitable for use	cum	127.05
M-090	Curing compound	litre	65.84
M-091	Delineators from ISI certified firm as per the standard drawing given in IRC - 79	each	1,023.00
M-092	Earth Cost or compensation for earth taken from private land	cum	66.55
M-093	Elastomeric slab seal expansion joint assembly manufactured by using chloroprene, elastomer for elastomeric slab unit conforming to clause 915.1 of IRC: 83 (part II),	metre	13,543.53
M-094	Electric Detonators @ 1 detonator for 1/2 gelatin stick of 125 gms each	100 nos	330.00
M-095	Epoxy compound with accessories for preparing epoxy mortar	kg	609.95
M-096	Epoxy mortar	kg	499.07
M-097	Epoxy primer	kg	438.35
M-098	Epoxy resin-hardner mix for prime coat	kg	367.07
M-099	Flag of red color cloth 600 x 600 mm	each	72.60
M-100	Flowering Plants	each	18.70
M-101	Galvanised MS flat clamp	nos	20.79
M-102	Galvanised steel wire crates of mesh size 100 mm x 100 mm woven with 4mm dia. GI wire in rolls of required size.	sqm	401.23
M-103	Galvanised structural steel plate 200 mm wide, 6 mm thick, 24 m long	kg	82.45
M-104	Gelatin 80%	kg	335.39
M-105	Geo grids	sqm	29.04
M-106	Geomembrane	sqm	35.09
M-107	Geonets	sqm	41.14
M-108	Geotextile	sqm	27.50
M-109	Geotextile filter fabric	sqm	90.09
M-110	GI bolt 10 mm Dia	nos	20.79
M-111	Grouting pump with agitator	hour	387.20
M-112	Grass (Doob)	kg	77.00
M-113	Grass (Fine)	kg	96.80
M-114	HDPE pipes 75mm dia	metre	422.40
M-115	HDPE pipes 90mm dia	metre	580.80
M-116	Hedge plants	each	12.10
M-117	Helical pipes 600mm diameter	metre	1,522.40
M-118	Hot applied thermoplastic compound	litre	860.75
M-119	HTS strand	tonne	85,391.11
M-120	Joint Sealant Compound	kg	612.11
M-121	Jute netting, open weave, 2.5 cm square opening for seedling and Mulching	sqm	660.00
M-122	LDO for steam curing	litre	126.83
M-123	M.S. Clamps	nos	16.94
M-124	M.S. Clamps	kg	85.25
M-125	M.S.shoes @ 35 Kg per pile of 15 m	kg	78.38
M-126	Mild Steel bars	tonne	58,410.00
M-127	Modular strip/box seal expansion joint including anchorage catering to a horizontal movement beyond 70 mm and upto 140mm assembly comprising of edge beams, central beam,2 modules chloroprene seal, anchorage elements, support and control system, all steel sections protected against corrosion and installed by the manufacturer or his authorised representative	metre	33,550.00

CHAPTER-1 CARRIAGE OF MATERIALS						
Sr No	Description	Unit	Quantity	Rate Rs	Cost Rs	Remarks/ Input ref.
1.1	Loading and Unloading of Stone Boulder/Stone aggregates/Sand/Kanker/Moorum.	cum				
	Placing tipper at loading point, loading with front end loader, dumping, turning for return trip, excluding time for haulage and return trip					
	Unit = cum					
	Taking output = 5.5 cum					
	Time required for					
	i) Positioning of tipper at loading point		1 Min			
	ii) Loading by front end loader 1 cum bucket capacity @ 25 cum per hour		13 Min			
	iii) Maneuvering, reversing, dumping and turning for return		2 Min			
	iv) Waiting time, unforeseen contingencies etc		4 Min			
	Total		20 Min			
	a) Machinery					
	Tipper 5.5 tonnes capacity	hour	0.33	701.80	231.59	P&M-048
	Front end-loader 1 cum bucket capacity @ 25 cum/hour	hour	0.33	1016.40	335.41	P&M-017
	b) Overhead charges @ 8 % on (a)				45.36	
	c) Contractor's profit @ 15 % on (a+b)				91.85	
	Cost for 5.5 cum = a+b+c				704.22	
	Rate per cum = (a+b+c)/ 5.5				128.04	
	Note Unloading will be by tipping.			say	128.00	
1.2	Loading and Unloading of Boulders by Manual Means					
	Unit = cum					
	Taking output = 5.5 cum					
	a) Labour					
	Mate	day	0.11	440.00	48.40	L-12
	Mazdoor for loading and unloading	day	0.75	302.50	226.88	L-13
	b) Machinery	day	0.75	701.80	526.35	P&M-048
	c) Overhead charges @ 8 % on (a+b)				64.13	
	d) Contractor's profit @ 15 % on (a+b+c)				129.86	
	Cost for 5.5 cum = a+b+c+d				995.62	
	Rate per cum = (a+b+c+d)/5.5				181.02	
	Note Unloading will be by tipping.			say	181.00	
1.3	Loading and Unloading of Cement or Steel by Manual Means and Stacking.					
	Unit = tonne					
	Taking output = 10 tonnes					
	a) Labour					
	Mate	day	0.08	440.00	35.20	L-12
	Mazdoor for loading and unloading	day	2.00	302.50	605.00	L-13
	b) Machinery					
	Truck 10 tonne capacity	hour	2.00	748.00	1496.00	P&M-057
	c) Overhead charges @ 8 % on (a+b)				170.90	
	d) Contractor's profit @ 15 % on (a+b+c)				346.06	
	Cost for 10 tonnes = a+b+c+d				2653.16	
	Rate per tonnes = (a+b+c+d)/10				265.32	
				say	265.00	
1.4	Cost of Haulage Excluding Loading and Unloading					
	Haulage of materials by tipper excluding cost of loading, unloading and stacking.					
	Unit = t.km					
	Taking output 10 tonnes load and lead 10 km = 100 t.km					
(I)	Surfaced Road					
	Speed with load : 25 km / hour.					
	Speed while Returning empty : 35 km / hour.					
	a) Machinery.					
	Tipper 10 tonne capacity					
	Time taken for onward haulage with load	hour	0.40	701.80	280.72	P&M-048
	Time taken for empty return trip.	hour	0.29	701.80	203.52	P&M-048
	b) Overhead charges @ 8 % on (a)				38.74	
	c) Contractor's profit @ 15 % on (a+b)				78.45	
	cost for 100 t km = a+b+c				601.43	
	Rate per t.km = (a+b+c)/100				6.01	
				say	6.00	
1.4	(II) Unsurfaced Graveled Road					
	Speed with load: 20 km / hour					
	Speed for empty return trip : 30 km / hour					
	a) Machinery					

CHAPTER-2 SITE CLEARANCE							
Sr No		Description	Unit	Quantity	Rate Rs	Cost Rs	Remarks/ Input ref.
2.1		Cutting of Trees, including cutting of Trunks, Branches and Removal					
		Cutting of trees, including cutting of trunks, branches and removal of stumps, roots, stacking of serviceable material with all lifts and up to a lead of 1000 metres and earth filling in the depression/nit					
		Unit = Each					
	(i)	Girth from 300 mm to 600 mm					
		a) Labour					
		Mate	day	0.02	440.00	8.80	L-12
		Mazdoors for cutting trees including cutting, refilling, compaction of backfilling and stacking of serviceable materials within 1000 metres lead by manual means.	day	0.60	302.50	181.50	L-13
		b) Machinery					
		Tractor-trolley	hour	0.10	402.60	40.26	P&M-053
		c) Overhead charges @ 8 % on (a+b)				18.44	
		d) Contractor's profit @ 15 % on (a+b+c)				37.35	
		Rate for each tree = a+b+c+d				286.36	
					say	286.00	
2.1	(ii)	Girth from 600 mm to 900 mm					
		a) Labour					
		Mate	day	0.04	440.00	17.60	L-12
		Mazdoors for cutting trees including cutting, refilling, compaction of backfilling, and stacking of serviceable materials within 1000 metres lead by manual means	day	0.90	302.50	272.25	L-13
		b) Machinery					
		Tractor-trolley	hour	0.30	402.60	120.78	P&M-053
		c) Overhead charges @ 8 % on (a+b)				32.85	
		d) Contractor's profit @ 15 % on (a+b+c)				66.52	
		Rate for each tree = a+b+c+d				510.00	
					say	510.00	
2.1	(iii)	Girth from 900 mm to 1800 mm					
		a) Labour					
		Mate	day	0.08	440.00	35.20	L-12
		Mazdoors for cutting trees including cutting, refilling, compaction of backfilling and stacking of serviceable materials within 1000 metres	day	2.00	302.50	605.00	L-13
		b) Machinery					
		Tractor-trolley	hour	0.40	402.60	161.04	P&M-053
		c) Overhead charges @ 8 % on (a+b)				64.10	
		d) Contractor's profit @ 15 % on (a+b+c)				129.80	
		Rate for each tree = a+b+c+d				995.14	
					say	995.00	
2.1	(iv)	Girth from 1800 mm to 2700mm					
		a) Labour					
		Mate	day	0.16	440.00	70.40	L-12
		Mazdoors for cutting trees including cutting, refilling, compaction of backfilling and stacking of serviceable materials within 1000 metres	day	4.00	302.50	1210.00	L-13
		b) Machinery					
		Tractor-trolley	hour	0.60	402.60	241.56	P&M-053
		c) Overhead charges @ 8 % on (a+b)				121.76	
		d) Contractor's profit @ 15 % on (a+b+c)				246.56	
		Rate for each tree = a+b+c+d				1890.27	
					say	1890.00	

CHAPTER-13 SUB-STRUCTURE						
Sr No	Description	Unit	Quantity	Rate Rs	Cost Rs	Remarks/ Input ref.
	Plate compactor/power rammer	hour	2.50	205.59	513.98	P&M-086
	Water Tanker	hour	0.05	748.00	37.40	P&M-060
	d) Overhead charges @ 20 % on (a+b+c)				5319.11	
	e) Contractor's profit @ 15 % on (a+b+c+d)				4787.20	
	Cost for 10 cum of granular backfill = a+b+c+d+e				36701.88	
	Rate per cum = (a+b+c+d+e)/10				3670.19	
				say	3670.00	
13.10	Providing and laying of Filter media with granular materials/stone crushed aggregates satisfying the requirements laid down in clause 2504.2.2. of MoRTH specifications to a thickness of not less than 600 mm with smaller size towards the soil and bigger size towards the wall and provided over the entire surface behind abutment, wing wall and return wall to the full height compacted to a firm condition complete as per drawing and Technical Specification.					
	Unit = cum					
	Taking output = 10 cum.					
	a) Labour					
	Mate	day	0.32	440.00	140.80	L-12
	Mazdoor for filling, watering, ramming etc.	day	7.00	302.50	2117.50	L-13
	Mazdoor (Skilled)	day	1.00	357.50	357.50	L-15
	b) Material					
	Filter media of stone aggregate conforming to clause 2504.2.2. of MoRTH specifications.	cum	12.00	1915.10	22981.20	M-012
	c) Machinery					
	Water Tanker of 6 KL capacity	hour	0.06	748.00	44.88	P&M-060
	d) Overhead charges @ 20 % on (a+b+c)				5128.38	
	e) Contractor's profit @ 15 % on (a+b+c+d)				4615.54	
	cost for 10 cum of Fiter Media = a+b+c+d+e				35385.79	
	Rate per cum = (a+b+c+d+e)/10				3538.58	
				say	3539.00	
13.14	Supplying, fitting and fixing in position true to line and level elastomeric bearing conforming to IRC: 83 (Part-II) section IX and clause 2005 of MoRTH specifications complete including all accessories as per drawing and Technical Specifications.					
	Unit: one cubic centimetre					
	Considering an elastomeric bearing of size 500 x 400 x 96 mm for this analysis.					
	Overall volume - 19200 cu.cm					
	Volume of 6 nos. 488 x 388 x 4 mm size reinforcing steel plates = 4545 cu.cm.					
	Hence volume of elastometer = 14655 cu.cm.					
	a) Labour					
	Mate	day	0.06	440.00	26.40	L-12
	Mazdoor	day	1.00	302.50	302.50	L-13
	Mazdoor (Skilled)	day	0.50	357.50	178.75	L-15
	b) Material					
	Elastomeric bearing assembly consisting of 7 layers of elastomer bonded to 6 nos. internal reinforcing steel laminates by the process of vulcanisation, complete with all components as per drawing and Technical Specifications.	each.	1.00	95711.00	95711.00	M-066
	Add 1 per cent of cost of bearing assembly for foundation anchorage bolts and consumables.				957.11	
	c) Overhead charges @ 20 % on (a+b)				19435.15	
	d) Contractor's profit @ 15 % on (a+b+c)				17491.64	
	cost for 19200cc of elastomeric bearing = a+b+c+d				134102.55	
	Rate per cc of elastomeric bearing = (a+b+c+d)/19200				6.98	
				say	7.00	

CHAPTER 16: REPAIR AND REHABILITATION						
Sr No	Description	Unit	Quantity	Rate Rs	Cost Rs	Remarks/ Input ref.
16.10	Applying pre-packed cement based polymer mortar of strength 45 Mpa at 28 days for replacement of spalled concrete Unit = sqm Taking output = 10 sqm Assumed thickness - 10 mm					
	a) Material					
	Acrylic polymer bonding coat	Litre	1.40	473.55	662.97	M-057
	pre-packed cement based polymer mortar of strength 45 Mpa at 28 days	kg	12.00	86.90	1042.80	M-145
	Add 3 per cent of (a) above for wastage.				51.17	
	b) Labour					
	Mate	day	0.04	440.00	17.60	L-12
	Mazdoor (Skilled)	day	0.50	357.50	178.75	L-15
	Mazdoor	day	0.50	302.50	151.25	L-13
	c) Overhead charges @ 10 % on (a+b)				210.45	
	d) Contractor's profit @ 15 % on (a+b+c)				347.25	
	Cost for 10 sqm = a+b+c+d				2662.25	
	Rate per sqm = (a+b+c+d)/10				266.22	
				say	266.00	
16.11	Epoxy bonding of new concrete to old concrete Unit = sqm Taking output = 10 sqm					
	a) Material					
	Epoxy resin with pot life not less than 60-90 minutes and satisfying testing as per clause 2803.9	kg	8.00	367.07	2936.56	M-098
	Add 3 per cent of (a) above for wastage.				88.10	
	b) Labour					
	Mate	day	0.04	440.00	17.60	L-12
	Mazdoor (Skilled)	day	0.50	357.50	178.75	L-15
	Mazdoor	day	0.50	302.50	151.25	L-13
	c) Overhead charges @ 10 % on (a+b)				337.23	
	d) Contractor's profit @ 15 % on (a+b+c)				556.42	
	Cost for 10 sqm = a+b+c+d				4265.90	
	Rate per sqm = (a+b+c+d)/10				426.59	
				say	427.00	
16.2	Replacement of Expansion Joints complete as per drawings Unit - 1 RM Taking output = 12 RM					
	a) Material					
	Epoxy for bonding new concrete to old concrete @ 0.8 kg/sqm	kg	9.60	609.95	5855.52	M-095
	M-30 grade cement concrete excluding OH & CP (Rate as per Items 14.1 C (i))	cum	3.60	11695.00	42102.00	Item 14.1(C)
	b) Labour					
	Removal of old expansion joint Including breaking of concrete, cutting of lugs and shifting of broken material etc.					
	Mate	day	0.26	440.00	114.40	L-12
	Mazdoor	day	6.00	302.50	1815.00	L-13
	Mazdoor (Skilled)	day	0.50	357.50	178.75	L-15
	c) Overhead charges @ 10 % on (a+b)				5006.57	
	d) Contractor's profit @ 15 % on (a+b+c)				8260.84	
	Cost for replacement of 12 RM = a+b+c+d				63333.07	
	Rate per RM = (a+b+c+d)/12				5277.76	
				say	5278.00	
Note	The rate for the installation of new expansion joints may be taken from the chapter on superstructure. Broken concrete will have to be replaced which has been					

INPUT RATES OF PLANT, MACHINERY, LABOUR AND MATERIALS
(A) Usage Rates of Plant and Machinery

Sl. No.	Description of Machine	Activity	Output of Machine	Output	Unit	Rate (Rs.)
P&M-001	Air Compressor	General Purpose	capacity In cfm	170/250	hour	405.90
P&M-002	Batching and Mixing Plant (a) 30 cum capacity	Concrete Mixing	cum/hour	20	hour	2,700.50
P&M-003	Batching and Mixing Plant (b) 15 - 20 cum capacity	Concrete Mixing	cum/hour	13	hour	2,249.50
P&M-004	Bitumen Pressure Distributor	Applying bitumen tack coat	sqm/hour	1750	hour	1,296.90
P&M-005	Bitumen Boiler oil fired	Bitumen Spraying	capacity In litre	1500	hour	397.10
P&M-006	Concrete Paver Finisher with 40 HP Motor	Paving of concrete surface	cum / hour	20	hour	3,907.20
P&M-007	Concrete Pump of 45 & 30 cum capacity	Pumping of concrete	cum / hour	33 / 22	hour	526.90
P&M-008	Concrete Bucket	For Pouring concrete	capacity In cum	1	hour	27.50
P&M-009	Concrete Mixer (a) 0.4/0.28 cum	Concrete Mixing	cum/hour	2.5	hour	591.80
P&M-010	Concrete Mixer (b) 1 cum	Concrete Mixing	cum/hour	7.5	hour	940.50
P&M-011	Crane (a) 80 tonnes	Lifting Purpose			hour	2,211.00
P&M-012	Cranes b) 35 tonnes	Lifting Purpose			hour	1,474.00
P&M-013	Cranes c) 3 tonnes	Lifting Purpose			hour	502.70
P&M-014	Dozer D - 80 - A 12	Spreading /Cutting / Clearing	cum/hour	300/ 150/250	hour	4,501.20
P&M-015	Dozer D - 50 - A 15	Spreading /Cutting / Clearing	cum/hour	200/ 120/150	hour	3,177.90
P&M-016	Emulsion Pressure Distributor	Applying emulsion tack coat	sqm/hour	1750	hour	1,013.10
P&M-017	Front End loader 1 cum bucket capacity	Soil loading / Aggregate loading	cum/hour	60 /25	hour	1,161.60
P&M-018	Generator (a) 125 KVA	Generation of electric Energy	KVA	100	hour	884.40
P&M-019	Generator(b) 63 KVA	Generation of electric Energy	KVA	50	hour	654.50
P&M-020	GSB Plant 50 cum	Producing GSB	cum/hour	40	hour	1,317.80
P&M-021	Hotmix Plant - 120 TPH capacity	DBM/BM/SDC/ Premix	cum/hour	40	hour	28,316.20
P&M-022	Hotmix Plant - 100 TPH capacity	DBM/BM/SDC/ Premix	cum/hour	30	hour	25,925.90
P&M-023	Hotmix Plant - 60 to 90 TPH capacity	DBM/BM/SDC/ Premix	cum/hour	25	hour	20,733.90
P&M-024	Hotmix Plant - 40 to 60 TPH capacity	DBM/BM/SDC/ Premix	cum/hour	17	hour	16,601.20
P&M-025	Hydraulic Chip Spreader	Surface Dressing	sqm/hour	1500	hour	3,187.80
P&M-026	Hydraulic Excavator of 1 cum bucket	Soil Ordinary/ Soil Marshy/ Soil Unsuitable	cum/hour	60 /60 /60	hour	1,800.70
P&M-027	Integrated Stone Crusher 100THP	Crushing of Spalls	TPH	100	hour	10,481.90
P&M-028	Integrated Stone Crusher 200 HP	Crushing of Spalls	TPH	200	hour	21,002.30
P&M-029	Kerb Casting Machine	Kerb Making	Rm/hour	80	hour	464.20
P&M-030	Mastic Cooker	Mastic Wearing coat	capacity In tonne	1	hour	71.64
P&M-031	Mechanical Broom Hydraulic	Surface Cleaning	sqm/hour	1250	hour	414.70
P&M-032	Motor Grader 3.35 mtr blade	Clearing /Spreading /GSB /WBM	cum/hour	200/200/50/ 50	hour	2,897.40
P&M-033	Mobile slurry seal equipment	Mixing and laying slurry seal	sqm/hour	2700	hour	1,219.90
P&M-034	Paver Finisher Hydrostatic with sensor control 100 TPH	Paving of DBM/ BM/SDC/ Premix	cum/hour	40	hour	3,388.00
P&M-035	Paver Finisher Mechanical 100 TPH	Paving of WMM /Paving of DLC	cum/hour	40/30	hour	1,686.30
P&M-036	Piling Rig with Bantonite Pump	0.75 m dia to 1.2 m dia Boring attachment	Rm/hour	2 to 3	hour	6,611.00
P&M-037	Pneumatic Road Roller	Rolling of Asphalt Surface	cum/hour	25	hour	1,574.10
P&M-038	Pneumatic Sinking Plant	Pneumatic Sinking of wells	cum/hour	1.5 to 2.00	hour	5,044.60
P&M-039	Pot Hole Repair Machine	Repair of potholes	cum/hour	4	hour	1,095.60
P&M-040	Prestressing Jack with Pump & access	Stressing of steel wires/stands			hour	163.90
P&M-041	Ripper	Scarifying	cum/hour	60	hour	51.70
P&M-042	Rotavator	Scarifying	cum/hour	25	hour	59.40
P&M-043	Road marking machine	Road marking	Sqm/hour	100	hour	117.70
P&M-044	Smooth Wheeled Roller 8 tonne	Soil /BM Compaction	cum/hour	70/25	hour	635.80
P&M-045	Tandem Road Roller	Rolling of Asfalt Surface	cum/hour	30	hour	1,053.80
P&M-046	Tipper - 5 cum	Transportation of soil, GSB, WMM, Hotmix etc.	Capacity In cum	5.5	km	191.40
P&M-047	Tipper - 5 cum	Transportation of soil, GSB, WMM, Hotmix etc.	Capacity In cum	5.5	tonne.km	30.25
P&M-048	Tipper - 5 cum	Transportation of soil, GSB, WMM, Hotmix etc.	Capacity In cum	5.5	hour	701.80
P&M-049	Transit Mixer 4.0/4.5 cum	Transportation of Concrete Mix to site	cum/hour	4.5	hour	1,313.40

INPUT RATES OF PLANT, MACHINERY, LABOUR AND MATERIALS

M-128	Modular strip/box seal expansion joint catering to a horizontal movement beyond 140mm and upto 210mm box/box seal joint assembly containing 3 modules/cells and comprising of edge beams, two central beams, chloroprene seal, anchorage elements, support and control system, all steel sections protected against corrosion and installed by the manufacturer or his authorised representative	metre	55,000.00
M-129	Nipples 12mm	nos	14.30
M-130	Nuts and bolts	kg	85.42
M-131	Paint	litre	333.74
M-132	Pavement Marking Paint	litre	663.74
M-133	Paving Fabric	sqm	159.50
M-134	Perforated geosynthetic pipe 150 mm dia	metre	322.81
M-135	Perforated pipe of cement concrete, Internal dia 100 mm	metre	268.40
M-136	Pesticide	kg	510.40
M-137	Pipes 200 mm dia, 2.5 m long for drainage	metre	618.75
M-138	Plastic sheath, 1.25 mm thick for dowel bars	sqm	67.10
M-139	Plastic tubes 50 cm dia, 1.2 m high	nos	310.20
M-140	Polymer braids	metre	20.90
M-141	Pre moulded Joint filler, 25 mm thick for expansion joint.	sqm	496.10
M-142	Pre-coated stone chips of 13.2 mm nominal size	cum	4,158.00
M-143	Preformed continuous chloroprene elastomer or closed cell foam sealing element with high tear strength, vulcanised in a single operation for the full length of a joint to ensure water tightness.	metre	514.80
M-144	Pre-moulded asphalt filler board	sqm	819.50
M-145	Pre-packed cement based polymer concrete of strength 45 Mpa at 28 days	kg	93.50
M-146	Primer	kg	204.60
M-147	Quick setting compound	kg	68.20
M-148	Random Rubble Stone	cum	2,108.70
M-149	RCC Pipe NP 4 heavy duty non pressure pipe 1000 mm dia	metre	16,394.40
M-150	RCC Pipe NP 4 heavy duty non pressure pipe 1200 mm dia	metre	18,753.90
M-151	RCC Pipe NP 4 heavy duty non pressure pipe 300 mm dia	metre	9,678.90
M-151	RCC Pipe NP 3 heavy duty non pressure pipe 600 mm dia	metre	9,678.90
M-152	Reflectors/glass beads	kg	425.70
M-153	Reinforcement strips 60 mm wide 5 mm thick as per clause 3102. (Copper Strips)	metre	49.50
M-154	Reinforcement strips 60 mm wide 5 mm thick as per clause 3102. (Galvanised carbon steel strips)	metre	22.00
M-155	Reinforcement strips 60 mm wide 5 mm thick as per clause 3102. (Glass reinforced polymer/ fibre reinforced polymer/polymeric strips)	metre	61.60
M-156	Reinforcement strips 60 mm wide 5 mm thick as per clause 3102. (Stainless steel strips)	metre	31.90
M-157	Reinforcement strips 60 mm wide 5 mm thick as per clause 3102. Aluminium strips)	metre	37.40
M-158	Rivets	each	290.40
M-159	Sand bags (Cost of sand and Empty cement bag)	nos	248.93
M-160	Sapling 2 m high 25 mm dia	each	6.05
M-161	Scrap tyres of size 900 x 20	nos	389.40
M-162	Seeds	kg	432.30
M-163	Selected earth	cum	181.50
M-164	Separation Membrane of Impermeable plastic sheeting 125 micron thick	sqm	14.30
M-165	Sheathing duct	metre	17.60
M-166	Shrubs	each	12.10
M-167	Sludge / Farm yard manure @ 0.18 cum per 100 sqm at site of work for turfing	cum	159.50
M-168	Sodium vapour lamp	each	4,295.50
M-169	Square Rubble Coursed Stone	cum	3,183.25
M-170	Steel circular hollow pole of standard specification for street lighting to mount light at 5 m height above deck level	each	13,259.79
M-171	Steel circular hollow pole of standard specification for street lighting to mount light at 9 m height above road level	each	19,675.81
M-172	Steel drum 300 mm dia 1.2 m high/empty bitumen drum	nos	209.00
M-173	Steel helmet and cushion block on top of pile head during driving.	kg	105.60
M-174	Steel pipe 25 mm external dia as per IS:1239	metre	335.50
M-175	Steel pipe 50 mm external dia as per IS:1239	metre	660.00
M-176	Steel wire rope 20 mm	kg	234.37
M-177	Steel wire rope 40 mm	kg	355.37
M-178	Strip seal expansion joint (80 mm movement)	metre	27.50
	Strip seal expansion joint (50 mm movement)	metre	4,730.00
	Strip seal expansion joint (40 mm movement)	metre	4,730.00
M-179	Structural Steel	tonne	89,540.00
M-180	Super plasticizer admixture IS marked as per 9103-1999	kg	249.15
M-181	Synthetic Geogrids as per clause 3102.8 and approved design and specifications.	sqm	47.30
M-182	Through and bond stone	each	70.74
M-183	Tie rods 20mm diameter	nos	105.60
M-184	Tiles size 300 x 300 mm and 25 mm thick	each	99.00
M-185	Timber	cum	21,362.55
M-186	Traffic cones with 150 mm reflective sleeve	nos	1,095.60
M-187	Tube anchorage set complete with bearing plate, permanent wedges etc	nos	671.00
M-188	Unstaked lime	tonne	14,740.00
M-189	Water	KL	22.00
M-190	Water based cement paint	litre	253.00
M-191	Welded steel wire fabric	kg	102.30
M-192	Wire mesh 50mm x 50mm size of 3mm wire	kg	91.30
M-193	Wooden ballies 2" Dia for bracing	each	39.93
M-194	Wooden ballies 8" Dia and 9 m long	each	181.50
M-195	Wooden packing	cum	7,768.20
M-196	Wooden staff for fastening of flag 25 mm dia, one m long	each	12.10
Overheads and Contractors profit			
	Overheads for Road Works	8 %	
	Contractors profit for Road Works	15 %	
	Overheads for Bridge Works	20 %	
	Overheads for Bridge Works (Rehabilitation)	10 %	
	Contractors profit for Bridge Works	15 %	

CHAPTER - 3						
EARTH WORK, EROSION CONTROL AND DRAINAGE						
Sr No	Description	Unit	Quantity	Rate Rs	Cost Rs	Remarks/ Input ref.
	Mate	day	0.40	440.00	176.00	L-12
	Mazdoor	day	10.00	302.50	3025.00	L-13
	b) Machinery					
	Water tanker 6 KL capacity including watering for 3 months	hour	14.00	748.00	10472.00	P&M-060
	Tractor-trolley	hour	2.40	440.00	1056.00	P&M-053
	c) Material					
	Seeds	kg	3.60	430.10	1548.36	M-162
	Sludge/Farm yard manure @ 0.18 cum per 100 sqm	cum	0.43	154.00	66.22	M-167
	Bitumen Emulsion	litre	0.00	31.76	0.00	M-077
	Non Woven Colr mat blanket made with full biodegradable colr fibers reinforced with fine Polypropylene netting securely	sqm	264.00	660.00	174240.00	M-121
	Cost of water for 3 months	KL	84.00	22.00	1848.00	M-189
	d) Overhead charges @ 8 % on (a+b+c)				15394.53	
	e) Contractor's profit @ 15 % on (a+b+c+d)				31173.92	
	Cost for 240 sqm = a+b+c+d+e				239000.02	
	Rate per sqm = (a+b+c+d+e)/240				995.83	
				say	996.00	
3.27	Sub-Surface Drains with Perforated Pipe					
	Construction of subsurface drain with perforated pipe of 100 mm Internal diameter of metal/ asbestos cement/ cement concrete/PVC, closely jointed, perforations ranging from 3 mm to 6 mm depending upon size of material surrounding the pipe, with 150 mm bedding below the pipe and 300 mm cushion above the pipe, cross section of excavation 450 x 550 mm. Excavated material to be utilised In roadway at site.					
	Unit = metre					
	Taking output = 10 metres					
	a) Labour					
	Mate	day	0.04	440.00	17.60	L-12
	Mazdoor for excavation and back filling	day	2.00	302.50	605.00	L-13
	c) Material					
	Perforated geosynthetic pipe 150 mm dia	metre	10.00	316.17	3161.73	M-134
	d) Overhead charges @ 8 % on (a+b+c)				302.75	
	e) Contractor's profit @ 15 % on (a+b+c+d)				613.06	
	Cost for 10 metres = a+b+c+d+e				4700.14	
	Rate per metre = (a+b+c+d+e)/10				470.01	
				say	470.00	
	Note Type of pipe may be modified depending upon provision in design.					
3.28	Aggregate Sub-Surface Drains					
	Construction of aggregate sub surface drain 300 mm x 450 mm with aggregates conforming to table 300-4, excavated material to be utilised In roadway.					
	Unit = metre					
	Taking output = 10 metres					
	a) Labour					
	Mate	day	0.02	440.00	8.80	L-12
	Mazdoor for excavation and back filling with aggregates	day	1.50	302.50	453.75	L-13
	b) Material					
	Crushed stone as per table 300-3	cum	1.35	2099.90	2834.87	M-012
	c) Overhead charges @ 8 % on (a+b)				263.79	
	d) Contractor's profit @ 15 % on (a+b+c)				534.18	
	Cost for 10 metres = a+b+c+d				4095.39	
	Rate per metre = (a+b+c+d)/10				409.54	
				say	410.00	
3.32	EARTH WORK ON HILL ROAD					
	Excavation in Hill Area in Soil by Mechanical Means					
	Excavation In soll in hilly area by mechanical means Including cutting and trimming of side slopes and disposing of excavated earth with all lifts and lead upto 1000 metres.					
	Unit = cum					
	Taking output = 260 cum					
	a) Labour					
	Mate	day	0.24	440.00	105.60	L-12
	Mazdoor for trimming slopes and helping in excavation etc.	day	6.00	302.50	1815.00	L-13
	b) Machinery					
	Dozer 80 HP (D-80 A 12)@ 43.28 cum per hour	hour	6.00	4313.10	25878.60	P&M-014
	Front end loader	hour	6.00	1114.30	6685.80	P&M-017
	Tipper 5.5cum capacity, 4 trips per hour	hour	12.00	701.80	8421.60	P&M-048

CHAPTER-14 SUPER-STRUCTURE						
Sr No	Description	Unit	Quantity	Rate Rs	Cost Rs	Remarks/ Input ref.
14.24	Modular Strip / Box Seal Joint					
	Providing and laying of a modular strip box seal expansion joint catering to a horizontal movement beyond 140mm and upto 210mm, complete as per approved drawings and standard specifications to be Installed by the manufacturer / supplier or their authorised representative ensuring compliance to the manufacturer's Instructions for installation.					
	Unit = Running meter					
	Taking output = 12 m					
	a) Labour					
	Mate	day	0.07	440.00	30.80	L-12
	Mazdoor	day	1.25	302.50	378.13	L-13
	Mazdoor (Skilled)	day	0.50	357.50	178.75	L-15
	b) Material					
	Supply of a modular box/box seal joint assembly containing 3 modules/cells and comprising of edge beams, two central beams, chloroprene seal, anchorage elements, support and control system, all steel sections protected against corrosion and installed by the manufacturer or his authorised	metre	12.00	55000.00	660000.00	M-128
	c) Overhead charges @ 20 % on (a+b)				132117.54	
	d) Contractor's profit @ 15 % on (a+b+c)				118905.78	
	Cost for 12 m Modular strip/box seal joint = (a+b+c+d)				911610.99	
	Rate per m = (a+b+c+d)/12				75967.58	
				say	75968.00	
	Note					
	1. The Installation shall be done by the manufacturer or his authorised representative to the satisfaction of the Engineer.					
	2. The concreting for joining the expansion joint assembly with the deck has not been Included In this analysis as the same is catered In the quantities of RCC deck.					
	3. The anchoring bars of the expansion joint assembly shall be welded to the main reinforcement of the deck.					
14.25	i) Providing, Fabrication, erection and launching of steel girder for steel composite superstructure including railing etc and fixing the girder with bearings complete with contractor's own steel of grade E250, quality B0, Killed and Normalised conforming to IS-2062 : 2011 including material, labour, tools and plants including all lead, lift complete as per approved drawings & Technical specification section 1000 and 1900 and as directed by Engineer. The fabrication cost includes all the types of battens, lacings, ties, stiffeners, packings, diaphragm, T&F bolts, HSFG bolts, rivets, shear studs, steel drifts, shop/field welds, templates, jigs, fixtures, back up supports, accessories etc. and marking each member for site identification known as "shipping marks" and transporting various components from fabrication yard (if workshop is not at the site) to bridge sites in packages, bundles and other means with due care and safe guards. The job also includes preparation of fabrication shop drawings and getting their approval from Engineer, fabrication of steel bridge including surface preparation by removing oil/grease using petroleum hydro carbon solvent conforming to IS : 1745, grit blasting to Sa 2-1/2 conforming to IS: 9954, (Swedish standard SIS-05-5900) primer coat by 2 pack organic ethyl zinc sillicate (min Dry Film Thickness (DFT) 70µm), intermediate coat by 2 pack epoxy micaceous iron oxide (min DFT 120µm) and finishing coat by Acrylic polyurethane (min DFT 50µm) all touchup works after erection and, inspection, tests on materials of welded fabricated parts as per specifications and directions of engineer. The cost also inclusive of assembly, erection and launching of Steel girders using contractor's own cranes of sufficient capacity and boom length as per site conditions or by other methods (The detailed scheme for erection and launching shall be prepared and got approved by contractor from Engineer). The erection cost includes all types of equipments required for erection by CC Cribbs of sufficient quantity and camber jacks, Trestle for supporting members of steel structures, mechanical and hydraulic jacks, steel wire ropes and winch crabs, launching nose of steel for cantilever launching, and any other equipment such as small cranes for field assembly, shifting of leafs of steel girders from horizontal position to vertical position and vice-versa, scaffolding, air compressors welding plants, pneumatic tools and mini workshop facilities at site etc.					

Summary of Unit Rates for Changtongya - Longleng Road

Item No.	Description	Unit	Adopted Unit Rate as per MORTH Data Book (Rs)	Item No. in Rate Analysis
1	2	3	4	5
	BILL NO. 1: GENERAL AND SITE CLEARANCE			
1.01	Clearing and Grubbing of Roadway as per Technical Specification Clause 201.	Ha	58066	2.3 (ii) A
1.02	Dismantling of Existing Structures or any Existing Civil Works within Right of Way, as per Technical Specification Clause 202.			
	(i) Brick Masonry	cum	362	2.4 (ii) B
	(ii) Stone Masonry	cum	362	2.4 (iii) B
	(iii) Lime/Plain Cement Concrete in all grades	cum	559	2.4 (i) II A
	(iv) Reinforced / Pre-stressed Cement Concrete in all grades	cum	938	2.4 (i) II B
	(v) Bituminous courses	cum	346	2.5 II A
	(vi) Non Bituminous courses	cum	586	2.5 I B
	(vii) Cement concrete Pavements	cum	1834	2.6
	(viii) Ordinary KM stone/Guard stone/Sign Post	No.	256	2.10 B
	(ix) Hectometre Stone	No.	51	2.10 C
	(x) 5th KM stone	No.	438	2.10 A
	(xi) Kerb stone	m	18	2.8
	(xii) All types of Fencing including posts, foundation concrete, back filling of pit etc. complete.	m	55	2.11
	(xiii) RCC Railing	m	261	16.18
	(xiv) Guard Rails/W Beam Crash Barrier	m	88	2.7
	(xv) Brick Pitching / Brick soling	cum	235	2.4 (ii) D
	(xvi) Removal of abandoned Telephone / Electric Poles and Lines including excavation and dismantling of foundation concrete and lines under the supervision of concerned department.	m	195	2.14
	(xvii) Removing all types of hume pipes and stacking serviceable material with all leads & lifts including earthwork and dismantling of masonry works.			
	(a) Up to 600 mm dia	m	206	2.4 (ix) A
	(b) Above 600 mm to 900 mm dia	m	279	2.4 (ix) B
	(c) Above 900 mm dia	m	478	2.4 (ix) C
	(xviii) Dismantling stone pitching/ dry stone spalls	cum	251	2.4 (iii) E
	(xix) Dismantling of CI Water Pipe Line up to 1000 mm dia.	m	147	2.12
1.03	Cutting of standing trees as per Technical Specification Clause 201			
	i) Girth from 300mm to 600mm	Each	59	2.1 (i) x 0.40 x 0.50
	ii) Girth above 600mm to 900mm	Each	106	2.1 (ii) x 0.40 x 0.50
	iii) Girth above 900mm to 1800mm	Each	205	2.1 (iii) x 0.40 x 0.50
	iv) Girth above 1800mm	Each	387	2.1 (iv) x 0.40 x 0.50
1.04	Removal of Stumps and roots as per Technical Specification Clause 201 and 305.			
	i) Girth from 300mm to 600mm	Each	176	2.1 (i) x 0.60
	ii) Girth above 600mm to 900mm	Each	319	2.1 (ii) x 0.60
	iii) Girth above 900mm to 1800mm	Each	614	2.1 (iii) x 0.60
	iv) Girth above 1800mm	Each	1160	2.1 (iv) x 0.60
	BILL NO. 2: EARTHWORK			
2.01	Excavation for Roadway including Drains in all types of soils and Ordinary Rocks as per Technical specifications Clause 301.	cum	262	(3.32+3.33)/2
2.02	Excavation for Roadway including Drains in Hard Rock (requiring blasting) as per Technical Specification Clause 301 and 302.	cum	427	3.34
2.03	Excavation for Roadway including Drains in Hard Rock (using controlled blasting) as per Technical Specification Clause 301 and 302.	cum	419	3.9
2.04	Excavation for Roadway including Drains in Hard Rock (blasting prohibited) as per Technical Specification Clause 301.	cum	562	3.8 A
2.05	Construction of Embankment complete as per Technical Specification Clause 305.	cum	248	3.16
2.06	Construction of Subgrade and Earthen Shoulders complete as per Technical Specification Clause 305.	cum	289	3.18
2.07	Preparation of Cut Formation as Subgrade satisfying designed CBR at 97% MDD complete as per Technical Specification Clause 301 and 305.	cum	75	3.19 Case-I
2.08	Preparation of Subgrade in Rocky Formation as per Technical Specification Clause 301.	sqm	314	4.12 x 0.100
	BILL NO 3: SUB-BASE & BASE COURSES			
3.01	Granular Sub Base complete as per Technical Specification Clause 401.	cum	3138	4.1 A (i)